A

Table of Contents
Key Event Title
Biological context
Cell Term
Organ Term
Event Components
Key Event Overview
AOPs Including This Key Event
Taxonomic Applicability
Life Stages
Sex Applicability
Key Event Description
How it is Measured or Detected
Evidence Supporting the Domain of Applicability
References

Endocytotic lysosomal uptake

Short name
Endocytosis

Explore in a Third Party Tool

Biological Context

Level of Biological Organization
Molecular

Cell term

Cell term
eukaryotic cell

Organ term

Event Components

Process	Object	Action
endocytosis	lysosomal membrane	increased

Key Event Overview

B

Table of Contents
Key Event Title
Biological context
Cell Term
Organ Term
Event Components
Key Event Overview
AOPs Including This Key Event
Taxonomic Applicability
Life Stages
Sex Applicability
Key Event Description
How it is Measured or Detected
Evidence Supporting the Domain of Applicability
References

Key Event Description

Endocytosis was discovered by the Belgian Nobel laureate Christian De Duve in 1963.

Endocytosis is a form of active transport in which molecules are transported into the cell by engulfing them in an energy-using process. In endocytosis, the material to be internalized is surrounded by an area of plasma membrane, which then buds off inside the cell to form a vesicle containing the ingested material. The ingestion of large particles (generally >250 nm in diameter) is termed phagocytosis (cell eating); phagocytosis is actin-dependent and restricted to professional phagocytes. The non-specific receptor-independent process to internalize fluids and solutes is called pinocytosis (cell drinking; via small vesicles of about 100 nm in diameter) and found in all cells (Cooper, 2000; Alberts et al., 2002; Oh and Park, 2014).

Receptor-mediated endocytosis can be clathrin-, and caveolin-dependent; these proteins mediate the invagination of the cell membrane.

The clathrin-mediated endocytotic pathway produces small (approx. 100 nm in diameter) vesicles coated with the cytosolic protein clathrin, forming clathrin-coated pits in the plasma membrane.

Caveolae-mediated endocytosis produces small (approximately. 50 nm in diameter) caveolae, flask-shape pits in the membrane coated with the protein caveolin, derived from lipid rafts (rigid membrane microdomains enriched with phospholipids, sphingolipids, and cholesterol). Clathrin- and caveolae-independent endocytosis is further sub-classified as Arf6- dependent, flotillin-dependent, Cdc42-dependent and RhoA-dependent endocytosis (Cleal et al., 2013; Villamil Giraldo et al., 2014; Iversen et al., 2011; Sahay et al., 2010; Kirkham and Parton, 2005; Mailaender and Landfester, 2009).

Vesicles rapidly lose their coats and fuse to form larger compartments, known as endosomes.

Early endosomes are the first compartment of the endocytic pathway and receive most types of vesicles coming from the cell surface. Endocytosed material is transferred to late endosomes and further to lysosomes, vacuoles of 1-2 µm in diameter containing hydrolytic enzymes in an acid milieu. Their main task is the degradation of ingested material (Villamil Giraldo et al., 2014).

Substances that are taken up selectively into lysosomes are called lysosomotropic agents (De Duve et al., 1974). These agents tend to have both lipophilic or amphiphilic compounds with basic moieties and accumulate in the acidic intracellular compartment, where they become protonated, therefore cannot diffuse back into the cytosol and accumulate within lysosomes, a phenomenon called "acid trapping" (Villamil Giraldo et al., 2014).

Although structurally and pharmacologically diverse, lysosomotropic compounds share certain